

Summary

Software developer with 4+ years of experience in building scalable **backend systems**, **DevOps automation**, and **AI/ML** integrations. Proficient in designing and deploying **microservices** using **Python** frameworks like **FastAPI** and **Django**, containerizing applications with **Docker**, orchestrating workloads with **Kubernetes**, and implementing **CI/CD** pipelines. Skilled in cloud platforms (**AWS**, **GCP**), monitoring, and **lightweight AI/ML** model integration for production-ready systems. Adept at bridging software **development**, **deployment**, and **ML** operations to deliver end-to-end solutions efficiently.

Core Skills:

- **Programming & Frameworks:** Python, FastAPI, Django, Flask, OOP, REST APIs, Microservices
- **DevOps & Cloud:** Docker, Kubernetes, Helm, CI/CD (Jenkins, GitHub Actions, GitLab CI), AWS (EC2, S3, IAM, EKS), GCP (GKE), Terraform, Infrastructure as Code
- **AI/ML & MLOps Touch:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, ML model deployment, MLflow, DVC, Joblib, Model versioning, Monitoring model drift
- **Databases & Data:** PostgreSQL, MySQL, SQLAlchemy, MongoDB, Redis
- **Monitoring & Logging:** Prometheus, Grafana, ELK, Python logging
- **Other Tools:** Git, GitHub, Linux, Bash scripting, Agile/Scrum

Education

- Masters in Computer Science from University Of Bridgeport

Certification

- AWS solutions architect- Associate

Professional Experience

Software Engineer | Black-Rock | May 2023 – Current

- Designed and developed a scalable backend system using **FastAPI** and **PostgreSQL**, implementing complete **CRUD** functionalities, **JWT**-based authentication, and **robust** unit and integration tests with **PyTest**, ensuring production-ready code quality.
- Containerized the backend using **Docker** and orchestrated deployment on **Kubernetes**, creating a multi-service architecture that includes an **AI-powered** sentiment analysis **microservice** integrated seamlessly into the Task Manager application.
- Built **CI/CD** pipelines using **GitHub** Actions to automate build, test, **containerization**, and deployment processes, reducing release cycles by 50% and enhancing developer productivity.
- Leveraged **AWS** cloud services, including **EC2**, **S3**, **IAM**, and **EKS** to deploy and scale the application securely and cost-effectively, ensuring high availability and **fault tolerance**.
- Implemented lightweight **AI/ML** features using **Scikit-learn**, exposing **ML models** via **FastAPI** endpoints, and tracked model versions and performance with **MLflow** and **Joblib** for maintainability and reproducibility.
- Monitored application performance and infrastructure health using **Prometheus** and **Grafana** dashboards, integrating alerting mechanisms for proactive issue resolution.
- Applied Infrastructure as Code (**Terraform**) to provision **AWS** resources automatically, enabling consistent environments across development, staging, and production, and reducing manual intervention errors.
- Implemented lightweight **MLOps** practices using **MLflow** and **DVC** for model versioning and experiment tracking, enabling retraining pipelines and continuous integration of **AI** features without impacting application stability.

Software Engineer | Mastercard | Aug 2020- Aug 2022

- Developed backend services using **Django** and **FastAPI** to manage product catalog, user accounts, and real-time analytics, supporting both **REST APIs** and asynchronous tasks for high throughput.
- Integrated **machine learning** features for product recommendation and sales forecasting using **Pandas**, **NumPy**, and **Scikit-learn**; exposed models via **API** endpoints, enabling personalized user experiences and data-driven decision-making.
- Containerized services with **Docker** and deployed on **Kubernetes** clusters on **AWS EKS**, implementing **Helm charts** for configuration management and horizontal pod scaling to ensure reliable system performance under variable loads.
- Automated **CI/CD** pipelines using Jenkins to handle build, testing, **containerization**, deployment, and rollback, improving deployment efficiency and reducing human errors.
- Employed monitoring solutions (**Prometheus**, **Grafana**) and centralized logging (**ELK stack**) to track **API** performance, system health, and **ML** model predictions in production environments, ensuring optimal uptime and reliability.
- Utilized **Terraform** for Infrastructure as Code, automating provisioning of **AWS EC2**, **S3**, **IAM** roles, and **RDS** instances to maintain consistency across environments while adhering to best security practices.
- Applied Infrastructure as Code (**Terraform**) to provision **AWS** resources automatically, enabling consistent environments across development, staging, and production, and reducing manual intervention errors.